

Dr. NAKUL TANWAR

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RESEARCH PROFILE

Bioinformatician with demonstrated research expertise in explainable AI for molecular disease characterization, omics-driven biomarker discovery, and reproducible computational pipelines. PhD research focused on integrating multi-modal molecular datasets to identify novel biomarkers in respiratory disorders through explainable machine learning. Strong publication record with peer-reviewed articles in high-impact journals. Experienced in designing rigorous computational workflows, mentoring junior researchers, and translating complex biological data into actionable clinical insights.

EDUCATION

Ph.D. Scholar, Department of Biotechnology

- Delhi Technological University | 2022–2026 (Awarded)
- Thesis Title: An Integrated Approach Towards the Identification of Novel Biomarkers in Respiratory Disorders
- Supervisor: Prof. Yasha Hasija
- Research Focus: Explainable AI for miRNA-mediated pathways in COPD, pulmonary fibrosis, and lung cancer; multi-omics integration; molecular mechanism elucidation

MSc Biotechnology

- Delhi Technological University | 2020–2022
- Project Title: Explaining Developmental Neurotoxicity By XAI
- Supervisor: Prof. Yasha Hasija

B.Sc. (Honors) Biotechnology

- Amity University, Gurugram | 2017–2020

PUBLICATIONS & RESEARCH CONTRIBUTIONS

A. Peer-Reviewed Journal Articles

- **Tanwar, N.,** Hasija, Y. (2025). "Explicating miRNA-mediated regulation of inflammatory pathways in COPD, MS, and lung cancer using explainable artificial intelligence: insights from peripheral blood profiles." *Integrative Biology*, 17, zyaf020. <https://doi.org/10.1093/intbio/zyaf020>
- **Tanwar, N.,** Hasija, Y. (2024). "Explicate molecular landscape of combined pulmonary fibrosis and emphysema through explainable artificial intelligence: a comprehensive analysis of ILD and COPD interactions using RNA from whole lung homogenates." *Medical & Biological Engineering & Computing*, 62, 2557–2570. <https://doi.org/10.1007/s11517-024-03099-8>
- **Tanwar, N.,** Hasija, Y. (2025). "An eXplainable AI and Conversational Decision Support System for Lung Cancer Diagnosis and Monitoring." *Technology and Health Care*. (Manuscript Under Review). <https://zenodo.org/records/15570514>

B. Book Chapter

- **Tanwar, N., &** Hasija, Y. (2024). "Overview of new trends on deep learning models for diabetes risk prediction." In *Internet of Things and Machine Learning for Type I and Type II Diabetes* (pp. 135-146). Elsevier.

C. IEEE Conference Presentations

- Yadav, K., **Tanwar, N., &** Hasija, Y. (2024). "Exploring the role of gut microbiota in hypertension: Insights from Machine Learning and eXplainable AI." 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT), June 2024.
- **Tanwar, N.,** Meena, J., & Hasija, Y. (2022). "Explicate Toxicity by eXplainable Artificial Intelligence." International Conference on Industry 4.0 Technology (I4Tech), September 2022.
- **Tanwar, N., &** Hasija, Y. (2022). "Explainable AI: Are We There Yet?" IEEE Delhi Section Conference (DELCON), February 2022.
- **Tanwar, N., &** Hasija, Y. (2022). "Deep Learning: A Tool in Biomedical Science." International Conference for Advancement in Technology (ICONAT), January 2022.

RESEARCH EXPERIENCE & TEACHING CONTRIBUTIONS

A. Teaching & Mentoring | Delhi Technological University | 2022–2025

- Mentored undergraduate and postgraduate students in bioinformatics research projects, providing guidance on experimental design, data analysis workflows, and ML model interpretation
- Co-designed and co-taught modules on explainable AI applications in biomedical sciences for B.Tech and M.Tech programs
- Supervised research documentation ensuring reproducibility, rigorous methodology, and publication-quality outputs
- Developed reproducible computational tutorials for omics data analysis and XAI model interpretation, enhancing student competency in modern bioinformatics

B. COVID-19 Molecular Diagnostics | Imperial Life Sciences | May–July 2021

- Validated RT-PCR assay kits for SARS-CoV-2 detection, optimizing reagent formulations and reaction conditions
- Established quality control protocols ensuring assay sensitivity, specificity, and reproducibility across batch production
- Contributed to rapid diagnostic tool development during pandemic response, supporting public health initiatives

C. Clinical Diagnostics & Pathology | Janakpuri Super Specialty Hospital | May–July 2019

- Performed comprehensive diagnostic analyses (CBC, blood smears, histopathology preparation)
- Mastered clinical workflow optimization, equipment maintenance, and quality assurance in diagnostic laboratory settings

CORE RESEARCH COMPETENCIES

A. Computational Omics Analysis

- Omics data integration (RNA-seq, miRNA profiling, whole lung homogenates)
- Statistical analysis, dimensionality reduction, pathway enrichment
- Hands-on expertise in differential expression analysis and molecular biomarker discovery

B. Machine Learning & Explainable AI (XAI)

- Supervised and unsupervised learning model development
- Feature engineering, model evaluation, and validation strategies
- Explainability frameworks: SHAP, LIME, attention mechanisms
- Deep learning architectures for biomedical applications

C. Large Language Models & Biomedical NLP

- LLM-augmented pipelines with XAI for biomedical text interpretation
- Prompt optimization for clinical decision support
- Conversational AI systems for disease diagnosis and monitoring
- Model evaluation and reliability assessment

D. Data Science & Reproducibility

- Statistical analysis and exploratory data analysis (EDA)
- Data cleaning and quality assurance
- Reproducible research practices and workflow documentation
- Python scientific computing (NumPy, Pandas, scikit-learn, Matplotlib)

E. Research Methodology

- Experimental design and hypothesis formulation
- Rigorous statistical testing and result interpretation
- Critical literature review and synthesis
- Peer review and publication processes

TECHNICAL SKILLS

- **Programming Languages:** Python (scientific computing, ML, automation)
- **ML Frameworks:** scikit-learn, TensorFlow/Keras, PyTorch, XGBoost
- **Data Analysis:** Statistical testing, EDA, dimensionality reduction, feature engineering
- **Bioinformatics Tools:** DESeq2, pathway analysis, gene set enrichment

- **LLM & NLP:** LLM prompt engineering, text classification, biomedical knowledge extraction
- **Data Visualization:** Matplotlib, Seaborn, Plotly, interactive dashboards
- **Version Control & Reproducibility:** Github, Jupyter Notebooks, documentation best practices

RECOGNITION & HONORS

- **Commendable Research Award**, Delhi Technological University (Research & Innovation Excellence Awards 2026) Recognition for outstanding contributions to explainable AI research in respiratory disease characterization
- **Commendable Research Award**, Delhi Technological University (Research & Innovation Excellence Awards 2025) Recognition for outstanding contributions to explainable AI research in respiratory disease characterization
- **GIAN Course Completion:** Medical Translational Research – Bench to Bedside (2025) Instructor: Prof. Hitesh Handa. ‘Advanced coursework in translating bench research to clinical applications’

INTELLECTUAL PROPERTY

- **Design Patent (Under Consideration)**
Title: Portable Food Spoilage Detection Device – Dual-sensor (gas + TDS/pH) system
Application No.: 461910-001
Status: FER Review (2025)